Let Φ_p be the characteristic function of singular binary cubic forms over $\mathbb{F}_p$. Then:

Theorem (Mori 2010)

We have

$$\widehat{\Phi_p}(v) = \begin{cases} p^{-1} + p^{-2} - p^{-3} & (v = 0), \\ p^{-2} - p^{-3} & (v \textit{ has splitting type } (1^3) \textit{ or } (1^2 1)), \\ -p^{-3} & (\textit{otherwise}). \end{cases}$$

Let $\Phi_{p^2} \pmod{p^2}$ be the characteristic function of nonmaximal cubic rings.

Theorem (Taniguchi-T., 2011)

We have

$$\widehat{\Phi_{p^2}}(v) = \begin{cases} p^{-2} + p^{-3} - p^{-5} & v/p : \textit{of type } (0), \\ p^{-3} - p^{-5} & v/p : \textit{of type } (1^3), (1^2 1), \\ -p^{-5} & v/p : \textit{of type } (111), (21), (3). \\ p^{-3} - p^{-5} & v : \textit{of type } (1^3_{**}), \\ -p^{-5} & v : \textit{of type } (1^3_{*}), (1^3_{\max}), \\ 0 & \textit{otherwise}. \end{cases}$$

Let $\Phi_{p^2} \pmod{p^2}$ be the characteristic function of nonmaximal cubic rings.

Theorem (Taniguchi-T., 2011)

We have

$$\widehat{\Phi_{p^2}}(v) = \begin{cases} p^{-2} + p^{-3} - p^{-5} & v/p : \textit{of type } (0), \\ p^{-3} - p^{-5} & v/p : \textit{of type } (1^3), (1^2 1), \\ -p^{-5} & v/p : \textit{of type } (111), (21), (3). \\ p^{-3} - p^{-5} & v : \textit{of type } (1^3_{**}), \\ -p^{-5} & v : \textit{of type } (1^3_{*}), (1^3_{\max}), \\ 0 & \textit{otherwise.} \end{cases}$$

So:

$$\frac{1}{p^8} \sum_{v \in V(\mathbb{Z}/p^2\mathbb{Z})} |\widehat{\Phi_{p^2}}(v)| \ll p^{-7}.$$

Let Φ_p be the characteristic function of singular binary quartic forms over $\mathbb{F}_p$. Then:

Theorem (Ishitsuka, Taniguchi, T., Xiao, 2024)

For a prime $p > 3$, we have

$$\widehat{\Phi_p}(v) = \begin{cases} p^{-1} + p^{-2} - p^{-3} & (v = 0), \\ p^{-2} - p^{-3} & (v \text{ has splitting type } (1^4) \text{ or } (1^3 1)), \\ \chi_{12}(p)(p^{-4} - p^{-3}) & (v \text{ has splitting type } (1^2 1^2)), \\ \chi_{12}(p)(p^{-4} + p^{-3}) & (v \text{ has splitting type } (2^2)), \\ \chi_{12}(p) p^{-4} & (v \text{ has splitting type } (1^2 11) \text{ or } (1^2 2)), \\ \chi_3(p) \left(\frac{I(v)}{p}\right) \cdot p^{-4} & (J(v) = 0, I(v) \neq 0), \\ a(E'_v) p^{-4} & (J(v) \neq 0, \mathrm{Disc}(v) \neq 0). \end{cases}$$

Here E'_v is the elliptic curve defined by

$$y^2 = x^3 - 3I(v)x^2 + J(v)^2,$$

with $a(E'_v) := p + 1 - \#E'_v(\mathbb{F}_p)$.

Let Ψ_p be the characteristic function of singular pairs of ternary quadratic forms.

Theorem (Taniguchi-T., 2016)

$$\widehat{\Psi_p}(x) = \begin{cases} p^{-1} + 2p^{-2} - p^{-3} - 2p^{-4} - p^{-5} + 2p^{-6} + p^{-7} - p^{-8} & x \in \mathcal{O}_{\mathbf{0}}, \\ p^{-3} - p^{-4} - 2p^{-5} + 2p^{-6} + p^{-7} - p^{-8} & x \in \mathcal{O}_{D\mathbf{1}^2}, \\ 2p^{-4} - 5p^{-5} + 3p^{-6} + p^{-7} - p^{-8} & x \in \mathcal{O}_{D\mathbf{11}}, \\ p^{-4} - 3p^{-5} + 2p^{-6} + p^{-7} - p^{-8} & x \in \mathcal{O}_{C\mathrm{s}}, \\ -p^{-5} + p^{-6} + p^{-7} - p^{-8} & x \in \mathcal{O}_{D\mathbf{2}}, \mathcal{O}_{D\mathrm{ns}}, \mathcal{O}_{C\mathrm{ns}}, \mathcal{O}_{B\mathbf{11}}, \mathcal{O}_{B\mathbf{2}}, \\ -p^{-6} + 2p^{-7} - p^{-8} & x \in \mathcal{O}_{\mathbf{1}^2\mathbf{1}^2}, \\ p^{-6} - p^{-8} & x \in \mathcal{O}_{\mathbf{2}^2}, \\ p^{-7} - p^{-8} & x \in \mathcal{O}_{\mathbf{1}^4}, \mathcal{O}_{\mathbf{1}^3\mathbf{1}}, \mathcal{O}_{\mathbf{1}^2\mathbf{11}}, \mathcal{O}_{\mathbf{1}^2\mathbf{2}}, \\ -p^{-8} & x \in \mathcal{O}_{\mathbf{1111}}, \mathcal{O}_{\mathbf{112}}, \mathcal{O}_{\mathbf{22}}, \mathcal{O}_{\mathbf{13}}, \mathcal{O}_{\mathbf{4}}. \end{cases}$$